

(a) Coalitions → unit directions

$$S_1 = \{1, 2\}$$

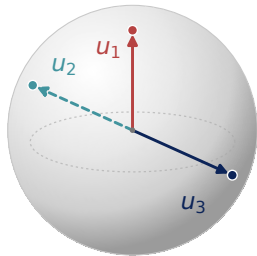
① ② ③ ④ $z_S = (1, 1, 0, 0)$ →

$$S_2 = \{1, 3\}$$

① ② ③ ④ $z_S = (1, 0, 1, 0)$ →

$$S_3 = \{2, 4\}$$

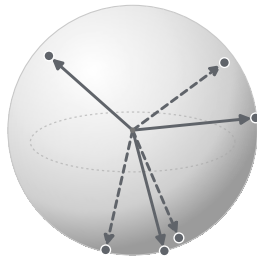
① ② ③ ④ $z_S = (0, 1, 0, 1)$ →



Center + normalize

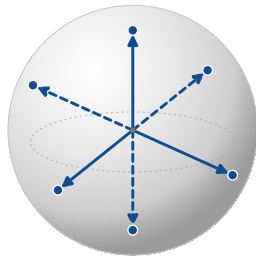
(b) Within-size sampling

IID



Random directions

INSIDE



Balanced directions